Integer-Tail Dynamics and a Pólya–Carlson Exclusion for Slow Modulations

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Abstract

We combine normalized integer reciprocal tails with analytic continuation of coefficient-interpolated generating functions. The resulting criterion excludes rational reciprocal sums for a class not treated by the preceding rational-profile note: for every $\alpha > 1$, $c \neq 0$, $\kappa > 0$, and $\eta > 0$, positive integers with

$$b_n/\alpha^{2^n} = 1 + c/(\log(n+2))^\kappa + O(e^{-\eta n})$$

have irrational reciprocal sum. Fractional-power profiles and nonzero periodic logarithmic profiles are also covered. The key analytic inputs are the classical Le Roy–Lindelöf continuation theorem and the Pólya–Carlson dichotomy. We separately derive the exact centered-residue dynamics behind Erdős Problems 243 and 263, an elementary half-integer obstruction at an integer base, and a finite-transient obstruction to purely local absorption arguments. These results do not settle the unrestricted $o(1)$ perturbation in Problem 263. Historical originality, external peer review, and proof-assistant verification are not claimed.

1 Target, results, and scope

The first question in Erdős Problem 263 asks whether

$$b_n \in \mathbb{Z}_{>0}, \qquad b_n/2^{2^n} \longrightarrow 1 \quad \Longrightarrow \quad \sum_n \frac{1}{b_n} \notin \mathbb{Q}. \tag{1}$$

The current community database and the consulted formal-conjecture declaration record this question as open [1, 2]. Its separate question about necessary growth for Type 2 irrationality sequences is not addressed here. Finitely many terms may always be changed or deleted: their reciprocal sum is rational.

Theorem 1.1 (Slow analytic modulations)**.** *Let $\alpha > 1$, $\nu \geq 2$, $\eta > 0$, and $c \in \mathbb{R} \setminus \{0\}$. Let (b_n) be positive integers, and suppose, for all sufficiently large n, that either*

$$\frac{b_n}{\alpha^{2^n}} = 1 + \frac{c}{(\log(n+\nu))^\kappa} + O(e^{-\eta n}), \qquad \kappa > 0, \tag{2}$$

or

$$\frac{b_n}{\alpha^{2^n}} = 1 + \frac{c}{(n+\nu)^\theta} + O(e^{-\eta n}), \qquad 0 < \theta < 1. \tag{3}$$

Then $\sum_n 1/b_n$ is irrational. The errors need not be monotone, continuous, or of one sign.

Corollary 1.2 (Rounded examples)**.** *For sufficiently large N, both series*

$$\sum_{n=N}^{\infty} \frac{1}{\lfloor 2^{2^n}(1 + 1/\log(n+2))\rfloor}, \qquad \sum_{n=N}^{\infty} \frac{1}{\lfloor 2^{2^n}(1 + 1/\sqrt{n+2})\rfloor}$$

are irrational. The base 2 *may be replaced by any real* $\alpha > 1$*. Floors may be replaced by ceilings, nearest-integer rounding, or a bounded integer change, provided the denominators remain positive.*

Proof. The relative rounding error is $O(\alpha^{-2^n})$, hence $O(e^{-\eta n})$ for every fixed $\eta > 0$. □

Theorem 1.3 (Periodic logarithmic modulations)**.** *Let* t_n *be a real periodic sequence, not identically zero, and let* $\alpha > 1$*,* $\nu \geq 2$*,* $\kappa > 0$*,* $\eta > 0$*. If positive integers satisfy*

$$\frac{b_n}{\alpha^{2^n}} = 1 + \frac{t_n}{(\log(n+\nu))^\kappa} + O(e^{-\eta n}),$$

then $\sum_n 1/b_n$ *is irrational.*

For example, $t_n = 1 + 2(-1)^n$ permits a leading correction that changes sign. No assertion is made for an arbitrary bounded, nonperiodic t_n.

Comparison with the preceding note. The previous rational-modulation manuscript allowed certain errors of polynomial order, such as $O(n^{-5})$ around $1 + 1/n$ [8]. The present note permits slower, nonrational leading profiles, but its positive-growth arguments require an exponentially small relative remainder. It is a complementary result, not a uniform improvement of the earlier error bound. In particular, $1 + 1/\log n + o(1/\log n)$ is *not* covered as a class.

2 Exact integer tails and the modular reformulation

The normalized tails provide the integer quantities used both in the new analytic argument and in the known pseudo-greedy reformulation of Problem 243 [3].

2.1 The integer certificate

Fix N and suppose $S_N = \sum_{n=N}^{\infty} 1/b_n = p/q \in \mathbb{Q}$, $q > 0$. Write

$$T_n = \sum_{j=n}^{\infty} \frac{1}{b_j}, \quad D_n = q \prod_{j=N}^{n-1} b_j, \quad C_n = D_n T_n. \tag{4}$$

Then

$$C_n = p \prod_{j=N}^{n-1} b_j - q \sum_{j=N}^{n-1} \frac{\prod_{k=N}^{n-1} b_k}{b_j} \in \mathbb{Z}_{>0}.$$

The fractions C_n/D_n are deliberately *not reduced*. Their exact evolution is

$$D_{n+1} = b_n D_n, \qquad C_{n+1} = b_n C_n - D_n. \tag{5}$$

Conversely, integrality at $n = N$ with some integer $q > 0$ already implies rationality of T_N. Thus no irrationality criterion is hidden in the normalization itself.

2.2 Centered residues and pseudo-greedy dynamics

Suppose $b_{n+1}/b_n^2 \to 1$. Quadratic growth gives

$$T_n = b_n^{-1} + b_{n+1}^{-1} + O(b_{n+1}^{-2}), \qquad T_n^{-1} = b_n - \frac{b_n^2}{b_{n+1}} + O(b_n^{-1}).$$

Consequently

$$\varepsilon_n = T_n^{-1} + 1 - b_n \longrightarrow 0. \tag{6}$$

For sufficiently large n, b_n is therefore the nearest integer to $T_n^{-1}+1$. This also follows from Koizumi's stronger quantitative theorem [3, Theorem 1].

For $C>0$ an integer, let $\mathrm{cent}_C(D)$ denote the unique integer congruent to D modulo C in $[-C/2, C/2)$. Once $|\varepsilon_n|<1/2$, put

$$e_n = \mathrm{cent}_{C_n}(D_n) = C_n\varepsilon_n.$$

Then the entire trajectory is deterministic:

$$\boxed{b_n = \frac{D_n - e_n}{C_n} + 1, \qquad C_{n+1} = C_n - e_n, \qquad D_{n+1} = b_n D_n.} \tag{7}$$

One further algebraic elimination yields

$$e_{n+1} = \mathrm{cent}_{C_n - e_n}(e_n b_n^2), \tag{8}$$

$$b_{n+1} = b_n(b_n-1) + 1 + \frac{e_n b_n^2 - e_{n+1}}{C_n - e_n}. \tag{9}$$

For instance, modulo C_{n+1} we have $C_n \equiv e_n$ and $D_n = C_n(b_n-1)+e_n \equiv e_n b_n$; multiplying by b_n proves (8). The quotient in (9) is an integer.

Proposition 2.1 (Absorption)**.** *If $e_n = 0$ at one step, then $e_j = 0$ for every $j \geq n$, and*

$$b_{j+1} = b_j^2 - b_j + 1 \qquad (j \geq n).$$

Proof. This follows immediately from (8) and (9). □

For a positive rational initial tail the nearest-integer rule defines a positive remainder at each step and these remainders decrease to zero. Indeed, $b_n > T_n^{-1}$; if the remainders had a positive limit, the denominators would be bounded, contradicting the repeated subtraction of their reciprocals. Thus the rule really gives a reciprocal expansion of the initial rational. Moreover, (7) gives

$$\frac{b_{n+1}}{b_n^2} = \frac{1 - b_n^{-1} + \varepsilon_n/b_n}{1-\varepsilon_n} + \frac{1-\varepsilon_{n+1}}{b_n^2}.$$

Hence $\varepsilon_n \to 0$ implies $b_{n+1}/b_n^2 \to 1$. This proves the following exact reformulation, also given by Koizumi:

Problem 243 is true $\iff$ every rational pseudo-greedy trajectory satisfying $e_n/C_n \to 0$ eventually has $e_n = 0$.

It is not a proof of that statement. The current community database records Problem 243 as open [1].

2.3 The growth-base constraint cannot be discarded

If $b_{n+1}/b_n^2 \to 1$ and $b_n \to \infty$, let $\delta_n = \log b_{n+1} - 2\log b_n = o(1)$. Then

$$\lambda = \lim_{n\to\infty} 2^{-n}\log b_n$$

exists, and

$$\log b_n - 2^n\lambda = -\sum_{j=n}^{\infty} 2^{n-j-1}\delta_j = o(1). \tag{10}$$

In particular $b_n \sim \alpha^{2^n}$ with $\alpha = e^{\lambda} > 1$. Positivity of λ follows, for large N, from $b_{n+1} \geq b_n^2/2$ and $b_N > 2$. A counterexample to (1) must have the special value $\lambda = \log 2$. A nonabsorbing rational trajectory at some other growth base would not automatically refute Problem 263.

3 An elementary half-integer obstruction

The implication from a positive solution of Problem 243 to the first question of Problem 263 can be seen without invoking any transcendence theorem for Sylvester growth constants.

Lemma 3.1 (Half-integer asymptotics)**.** *Suppose a positive integer sequence tends to infinity and eventually satisfies $b_{n+1} = b_n^2 - b_n + 1$. There is $\alpha > 1$ such that*

$$b_n = \alpha^{2^n} + \frac{1}{2} + O(\alpha^{-2^n}).$$

Consequently it cannot be asymptotic to a^{2^n} for an integer $a \geq 2$.

Proof. Set $y_n = b_n - 1/2$. Then exactly

$$y_{n+1} = y_n^2 + \frac{1}{4}, \qquad \log y_{n+1} - 2\log y_n = \log\left(1 + \frac{1}{4y_n^2}\right).$$

With $\log \alpha = \lim 2^{-n} \log y_n$, the same summation as in (10) gives

$$2^n \log \alpha - \log y_n = \sum_{j=n}^{\infty} 2^{n-j-1} \log\left(1 + \frac{1}{4y_j^2}\right) = O(y_n^{-2}).$$

Here $y_{n+1} \geq y_n^2$ bounds the tail. Exponentiating yields $\alpha^{2^n} = y_n + O(y_n^{-1})$, proving the asserted expansion. If also $b_n \sim a^{2^n}$, the growth base must be $\alpha = a$. The integer $b_n - a^{2^n}$ would then tend to $1/2$, an impossibility. □

Corollary 3.2. *A positive solution of Problem 243 implies a positive solution of the first question in Problem 263. This is an implication, not an equivalence.*

The same reasoning supplies useful necessary conditions on a possible counterexample to (1). In its eventual centered-residue trajectory, $e_n \neq 0$ for every n, since a zero residue would trigger the contradiction in Lemma 3.1. Because $e_n/C_n \to 0$ and $|e_n| \geq 1$, necessarily

$$C_n \longrightarrow \infty, \qquad \log C_n = o(n). \tag{11}$$

The second assertion also follows from $C_{n+1}/C_n = 1 - \varepsilon_n \to 1$. Writing $u_n = b_n/2^{2^n}$, the product formula below shows in addition

$$\prod_{j=N}^{n-1} u_j \longrightarrow \infty. \tag{12}$$

If the negative gap mass were finite, $\sum_n \max(-\varepsilon_n, 0) < \infty$, then $C_{n+1} = C_n(1 - \varepsilon_n)$ would give a uniform upper bound on C_n. Thus this mass must diverge for a counterexample. These are restrictions, not an exclusion of every possible counterexample.

4 An analytic-continuation exclusion criterion

The following criterion is the new mechanism used in this note. It separates the arithmetic issue (integer tails) from the analytic one (continuation of a comparison generating function).

Lemma 4.1 (Tail product formula)**.** *Let $A_n = \alpha^{2^n}$, $\alpha > 1$, and $b_n = A_n u_n$ with $u_n \to 1$. Then*

$$T_n = \frac{1}{b_n}\left(1 + O(A_n^{-1})\right).$$

Under the rationality assumption in (4),

$$C_{N+m} = \frac{q}{A_N} \frac{\prod_{j=N}^{N+m-1} u_j}{u_{N+m}} \left(1 + O(A_{N+m}^{-1})\right). \tag{13}$$

Proof. Since $b_{n+1} \asymp b_n^2$, the terms after b_n^{-1} sum to $O(b_n^{-2})$. Also $\prod_{j=N}^{N+m-1} A_j = A_{N+m}/A_N$. Substitution in (4) proves (13). □

Theorem 4.2 (Continuation versus integer growth)**.** *Let $R_n > 0$ tend to 1, and put*

$$H_m = \frac{\prod_{j=N}^{N+m-1} R_j}{R_{N+m}} \qquad (m \geq 0).$$

Assume

$$\log H_m = o(m), \qquad \frac{\log H_m}{\log m} \longrightarrow +\infty, \tag{14}$$

and that $F_H(z) = \sum_{m\geq 0} H_m z^m$ admits analytic continuation through at least one point of the unit circle. If positive integers satisfy

$$b_n/\alpha^{2^n} = R_n + O(e^{-\eta n}), \qquad \alpha > 1, \quad \eta > 0,$$

then $\sum_n 1/b_n$ is irrational.

Proof. Suppose otherwise and use the integer tails C_n. Write $b_n = A_n R_n v_n$, so $v_n = 1+O(e^{-\eta n})$. Their product converges to a positive limit V_∞, and

$$\frac{\prod_{j=N}^{N+m-1} v_j}{v_{N+m}} = V_\infty(1 + O(e^{-\eta m})).$$

By Lemma 4.1, for a constant $K > 0$,

$$C_{N+m} = KH_m + E_m, \qquad E_m = O(e^{-\eta m/2}). \tag{15}$$

The exponential bound uses $H_m = e^{o(m)}$ and the doubly exponential smallness of A_{N+m}^{-1}. Thus

$$F_C(z) := \sum_{m\geq 0} C_{N+m} z^m = KF_H(z) + F_E(z),$$

where F_E is holomorphic in a disk of radius greater than 1. The integer-coefficient series F_C has radius of convergence exactly 1 and extends through the same boundary point as F_H.

The Pólya–Carlson theorem says that an integer-coefficient series of radius 1 is rational or has the whole unit circle as a natural boundary; see [4, Theorem 1.1]. Hence F_C is rational. Partial fractions of a rational function with no poles inside the unit disk imply $C_{N+m} = O(m^M)$ for some fixed M. This contradicts (14) and (15). □

Remark 4.3 (A complementary elementary case). With the same summable multiplicative error, if instead $H_m \to 0$, then $C_{N+m} \sim KH_m \to 0$, directly contradicting positive integrality. No continuation theorem is needed in that case.

5 Building the analytic continuation explicitly

We now verify, rather than assume, the analytic hypothesis for the profiles in Theorem 1.1.

5.1 The classical continuation input

We use the following form of the Le Roy–Lindelöf theorem [5, Introduction]: if Φ is holomorphic in a neighborhood of the closed right half-plane and, for every $\epsilon > 0$,

$$|\Phi(z)| \leq C_\epsilon e^{\epsilon|z|} \qquad (\Re z \geq 0),$$

then $\sum_{m\geq 0} \Phi(m) w^m$ continues off the nonnegative real axis. Combined with its original unit disk, this supplies continuation to the slit plane $\mathbb{C} \setminus [1, \infty)$ when the radius is 1. This is the zero-exponential-type case of the classical theorem. It can also be obtained by the Lindelöf contour representation: the factor $\pi/\sin(\pi z)$ gives vertical exponential decay and the zero-type bound allows the residue contour to be shifted to the right.

5.2 An interpolation by convergent differences

For either profile, take N so large that all branches below are holomorphic for $\Re z > N-1$, and define

$$f(z) = \operatorname{Log} R(z),$$

where

$$R(z) = 1 + \frac{c}{(\operatorname{Log}(z+\nu))^{\kappa}} \quad \text{or} \quad R(z) = 1 + \frac{c}{(z+\nu)^{\theta}}.$$

All logarithms and powers are the principal branches on that half-plane; by increasing N we ensure $|R(z)-1| < 1/2$ there. Define

$$g(z) = \sum_{k=0}^{\infty} \bigl(f(N+k) - f(N+k+z)\bigr), \qquad \Phi(z) = \exp\bigl(g(z) - f(N+z)\bigr). \tag{16}$$

The series is locally uniformly convergent for $\Re z > -1/2$, because the following derivative bounds are summable along the positive integers:

	$\lvert f(w)\rvert$	$\lvert f'(w)\rvert$
logarithmic	$O((\log\lvert w+\nu\rvert)^{-\kappa})$	$O(\lvert w+\nu\rvert^{-1}(\log\lvert w+\nu\rvert)^{-\kappa-1})$
fractional power	$O(\lvert w+\nu\rvert^{-\theta})$	$O(\lvert w+\nu\rvert^{-1-\theta})$

(17)

For an integer $m \geq 0$, telescoping finite sums in (16) and letting the upper limit tend to infinity gives

$$g(m) = \sum_{j=N}^{N+m-1} f(j), \qquad \Phi(m) = H_m.$$

The omitted last m terms tend to zero for each fixed m.

It remains to check the growth in the entire right half-plane, not just on the positive axis. Write $r = |z| \geq 2$, $\Re z \geq -1/4$, and split the sum at $k = 2r$. For $k \leq 2r$, use the two separate bounds on f in (17); for $k > 2r$, integrate f' on the segment from $N+k$ to $N+k+z$. Uniformly in this half-plane one obtains

$$\begin{aligned}
|g(z)| &\leq C\left(1 + \sum_{k\leq 2r} \frac{1}{(\log(N+k))^{\kappa}} + r\sum_{k>2r} \frac{1}{(N+k)(\log(N+k))^{\kappa+1}}\right) \\
&= O\left(\frac{r}{(\log r)^{\kappa}}\right) \qquad \text{in the logarithmic case,} \\
|g(z)| &\leq C\left(1 + \sum_{k\leq 2r} (N+k)^{-\theta} + r\sum_{k>2r} (N+k)^{-1-\theta}\right) \\
&= O(r^{1-\theta}) \qquad \text{in the power case.}
\end{aligned}$$

The fixed shifts in these bounds are harmless. The same estimates, with an extra bounded term, bound $\log|\Phi(z)|$ from above. Both bounds are $o(r)$. Thus Φ has zero exponential type on a neighborhood of the closed right half-plane, as required. The generating function F_H therefore has the desired continuation.

5.3 Growth and proof of the main theorem

For $c > 0$, summing $f(n)$ gives

$$\log H_m = (c + o(1))\frac{m}{(\log m)^{\kappa}} \qquad \text{under (2)}, \tag{18}$$

$$\log H_m = \left(\frac{c}{1-\theta} + o(1)\right) m^{1-\theta} \qquad \text{under (3)}. \tag{19}$$

For example, $\log(1+x) = x + O(x^2)$ and the quadratic correction is $o(\sum x_n)$ for each of these slowly decaying positive sequences. The elementary estimates $\sum_{j\le m}(\log j)^{-\kappa} \sim m/(\log m)^{\kappa}$ and $\sum_{j\le m} j^{-\theta} \sim m^{1-\theta}/(1-\theta)$ complete the calculation. Both expressions satisfy (14). Theorem 4.2 now proves irrationality. For $c < 0$, the same asymptotics give $\log H_m \to -\infty$, and Remark 4.3 applies. This proves Theorem 1.1.

6 Periodic profiles: sign changes are allowed

Here we prove Theorem 1.3. Let p be a period and

$$\mu = \frac{1}{p}\sum_{r=0}^{p-1} t_{N+r}, \qquad v = \frac{1}{p}\sum_{r=0}^{p-1} t_{N+r}^2.$$

If $\mu \neq 0$, periodic summation and the expansion of $\log(1+x)$ give

$$\log H_m = (\mu + o(1))m/(\log m)^{\kappa}.$$

Thus $\mu < 0$ is handled by decay, while $\mu > 0$ has the required superpolynomial, subexponential growth.

For $\mu > 0$ one must interpolate residue classes separately; replacing t_n by a trigonometric function of a complex argument would not provide the required growth bound. Define, for $0 \le r < p$,

$$f_r(z) = \operatorname{Log}\left(1 + \frac{t_{N+r}}{(\operatorname{Log}(N+r+pz+\nu))^{\kappa}}\right), \qquad F(z) = \sum_{r=0}^{p-1} f_r(z).$$

Increasing N again makes all these functions holomorphic for $\Re z > -1/2$. Set

$$G(z) = \sum_{k=0}^{\infty}(F(k) - F(k+z)), \qquad \Phi_r(z) = \exp\left(G(z) + \sum_{j=0}^{r-1} f_j(z) - f_r(z)\right).$$

Then $\Phi_r(k) = H_{pk+r}$. The same prefix/tail estimates as above prove zero exponential type for every Φ_r. Hence each $\sum_{k\ge 0} H_{pk+r} w^k$ continues through every point of the unit circle except possibly 1. The decomposition

$$F_H(z) = \sum_{r=0}^{p-1} z^r \sum_{k\ge 0} H_{pk+r}(z^p)^k$$

gives continuation through every z with $|z| = 1$ and $z^p \neq 1$. Theorem 4.2 applies.

Finally suppose $\mu = 0$. Because t_n has bounded partial sums, $\sum t_n/(\log(n+\nu))^{\kappa}$ converges by summation by parts. Since t_n is not identically zero, $v > 0$, and the quadratic term in $\log(1+x)$ gives

$$\log H_m = -\left(\frac{v}{2} + o(1)\right)\frac{m}{(\log m)^{2\kappa}} \longrightarrow -\infty.$$

The cubic remainder is $o(m/(\log m)^{2\kappa})$. Remark 4.3 completes the proof.

7 Why finite local tests cannot close the absorption gap

The following construction illustrates a genuine obstacle. Related flat-residue constructions are presented in the independent working report [7]; the proof here is explicit and does not depend on an unverified claim from that report.

Proposition 7.1 (Arbitrarily long small nonzero gaps)**.** *Given $L \geq 1$ and $\epsilon > 0$, a positive rational initial tail can be chosen so that its centered-residue expansion has L consecutive nonzero gaps of magnitude less than ϵ, followed by a zero gap and absorption.*

Proof. Choose a prime $p > L + \max(3, \epsilon^{-1})$ and put

$$C = p - L, \qquad M = \prod_{j=0}^{L-1} (C + j).$$

Since $p \nmid M$, choose $T > 0$ with $MT \equiv 1 \pmod{p}$. Increasing T by a multiple of p, arrange $MT - 1 > 10C$. Start the unreduced trajectory at $C_0 = C$, $D_0 = MT - 1$.

Let $M_j = \prod_{i=j}^{L-1}(C + i)$. We show inductively that for $j < L$,

$$C_j = C + j, \quad D_j \equiv -1 \pmod{M_j}, \quad p \mid D_j.$$

These assertions hold at $j = 0$. Since $C_j \geq 3$, the centered residue is $e_j = -1$. Therefore $C_{j+1} = C_j + 1$. Also $b_j = (D_j + 1)/C_j + 1 \equiv 1 \pmod{M_{j+1}}$, whence $D_{j+1} = D_j b_j \equiv -1 \pmod{M_{j+1}}$. Divisibility by p persists. At step L, $C_L = p \mid D_L$, so $e_L = 0$ and absorption follows. The nonzero gaps are exactly $-1/(C + j)$, all of magnitude less than ϵ. The recurrence and the final telescoping Sylvester tail show that the reciprocal sum is the initial rational C_0/D_0. □

This is not a counterexample to either problem: the gaps eventually vanish, and the limiting growth base is not an integer by Lemma 3.1. It does rule out a uniform finite-window argument asserting that a fixed number of sufficiently small gaps forces an immediate zero gap. Treating centered residues as independent random samples would likewise be a heuristic, not a proof.

8 The remaining full-problem obstruction

For an unrestricted perturbation $b_n/2^{2^n} = 1 + o(1)$, the exact conditions proved above imply that any rational reciprocal sum would produce a rational initial state and a deterministic trajectory with

$$\boxed{e_n \neq 0, \quad e_n/C_n \longrightarrow 0, \quad \lim_{n\to\infty} 2^{-n} \log b_n = \log 2.} \tag{20}$$

Conversely, a rational pseudo-greedy trajectory satisfying these conditions would refute the first question in Problem 263, by (10). Thus exclusion of these trajectories is an exact remaining task.

The Pólya–Carlson theorem does not by itself exclude them. Their integer-tail generating function has radius 1, but may have a natural boundary. No analytic continuation follows merely from $C_{n+1}/C_n \to 1$. Nor have we proved that the rational-generating-function branch forces absorption in every case. In our explicit positive profiles, continuation excludes the natural-boundary branch and superpolynomial growth excludes the rational branch; both additional ingredients matter.

Similarly, the usual small-positive-integer argument at the unmodified tail level stalls at this critical growth scale:

$$\left(\prod_{j=N}^{n-1} b_j\right) T_n \asymp \prod_{j=N}^{n-1} (b_j/2^{2^j}).$$

The right side can grow rather than tend to zero. Fixed-order finite differences alone also do not make a profile of size $\exp(n/(\log n)^\kappa)$ tend to zero. The analytic continuation step is precisely what replaces that earlier approach, not a proof that every possible irregular profile is covered.

9 Provenance and verification record

Koizumi's published paper establishes the pseudo-greedy framework and proves that all but countably many real bases are Type 2 irrationality bases [3]. It does not identify base 2 as a nonexceptional base. Duverney's survey includes the classical integer-base criterion requiring summable relative perturbations [6, Theorem 2.3]. The leading positive perturbations $1/(\log n)^{\kappa}$ and $n^{-\theta}$, $0 < \theta < 1$, are not summable. Consequently that particular criterion cannot simply be substituted for the present proof. This comparison is not an exhaustive priority search.

The complex-analytic ingredients are classical and explicitly cited. The present note proves their hypotheses for the displayed profiles and derives the resulting exclusion criterion. It makes no claim that this application is historically first. No unproved result in the user's corpus or in the preceding research note is used as a premise.

The accompanying `verify_exact.py` uses only integer and rational arithmetic. It checked 10,000 centered-residue states and their two-step identities, the trajectory beginning at 11/29, flat transients of lengths 1 through 10, eight exact rounded square-root-profile denominators, and 1,000 instances of the half-shift identity. All checks passed in the saved run. These finite checks do not verify the complex-analytic arguments or prove an infinite irrationality theorem. No Lean or other proof-assistant verification, external peer review, or unrestricted solution of Problem 263 is asserted.

References

[1] T. Tao and contributors, *Erdős problem database*, entries 243 and 263, consulted 28 October 2026. https://github.com/teorth/erdosproblems.

[2] The Formal Conjectures Authors, *Erdős Problem 263*, `FormalConjectures/ErdosProblems/263.lean`, consulted 28 October 2026. Formal Conjectures repository. The main question is tagged `research open`; its `sorry` is a placeholder, not a proof.

[3] J. Koizumi, *Irrationality of the reciprocal sum of doubly exponential sequences*, Integers **26** (2026), A28. https://math.colgate.edu/~integers/aa28/aa28.pdf. Also arXiv:2504.05933.

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[7] *Erdős Problem a Day: Problem 243*, independent working report, dated 12 August 2026, consulted 28 October 2026. https://www.erdosproblemaday.com/report/243. Cited for prior flat-transient constructions, not as an independently refereed resolution of either problem.

[8] *A Rational-Modulation Exclusion for Doubly Exponential Reciprocal Sums*, research note prepared for Keio, 28 October 2026. Earlier manuscript in the present research session; not a proof dependency.